

Non-Regular Languages Proven Using the Pumping Lemma

Equal Numbers of a's and b's

Set Notation: $L = \{ a^n b^n \mid n \geq 0 \}$

Strings Produced: ϵ , ab, aabb, aaabbb, ...

Palindromes

Set Notation: $L = \{ w \mid w = w^R, \text{ where } w^R \text{ is the reverse of } w \}$

Strings Produced: ϵ , a, b, aa, bb, aba, abba, ...

More a's Than b's

Set Notation: $L = \{ w \mid \#a(w) > \#b(w) \}$

Strings Produced: a, aa, aaa..., aab, aaab, aaabb...

Equal Numbers of a's, b's, and c's

Set Notation: $L = \{ a^n b^n c^n \mid n \geq 0 \}$

Strings Produced: ϵ , abc, aabbcc, aaabbbccc, ...

Power of Two

Set Notation: $L = \{ a^{2^n} \mid n \geq 0 \}$

Strings Produced: a, aa, aaaa, ...

Strings with Prime Length

Set Notation: $L = \{ w \mid |w| \text{ is prime} \}$

Strings Produced: a, aa, aaa, aaaaa, ...

Language of Balanced Parentheses

Set Notation: $L = \{ w \mid w \text{ contains balanced parentheses} \}$

Strings Produced: ϵ , (), (()), ()(), ...

Unequal Numbers of a's and b's

Set Notation: $L = \{ w \mid \#a(w) \neq \#b(w) \}$

Strings Produced: a, b, aa, bb, baa, bbb, ...

Strings with $a^n b^m$, Where $n \neq m$

Set Notation: $L = \{ a^n b^m \mid n \neq m, n, m \geq 0 \}$

Strings Produced: a, b, aab, aaabb, bba, ...

Strings Over {a, b} Where All a's Appear Before b's

Set Notation: $L = \{ w \mid w = a^n b^m, n, m \geq 0 \text{ and no ba as a substring} \}$

Strings Produced: ϵ , a, b, aa, ab, aab, ...

Strings Where w Contains More b's Than a's

Set Notation: $L = \{ w \in \{a, b\}^* \mid \#b(w) > \#a(w) \}$

Strings Produced: b, bb, abb, baaabb, ...